Native-VC Brill Collapse: Outer-Boundary Convergence and N256 KO Scan

AthenaK numerical investigation

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Abstract

We isolate two hypotheses in the repaired vertex-centered Cartoon Z4c Brill calculation. First, production-matched radial constraint inventories and a new $R_{\text{out}} = 128M$ common-tree experiment show that the apparent N256–N512 flattening in the old global Figure-2 norms is an outer-boundary/shell effect. Second, an independent $R_{\text{out}} = 16M$ N256 scan shows that stronger Kreiss–Oliger (KO) dissipation strongly delays the late refinement cascade. The two conclusions are kept separate; no production default is changed.

1 Scope and numerical authority

All work starts from commit `953f2724c00a2efd2f9fad91ae9a784639954a3b`. The production VC transfer, Z4c equations, Cartoon regularization, gauge, constraint damping, and admissibility gates are unchanged. The boundary experiment changes only domain size, minimum SMR structure, and absolute level ceilings. The KO experiment changes only `z4c/diss`.

The Cartoon history diagnostic already uses

$$dV = 2\pi\rho\,\sqrt{\det\gamma}\,d\rho\,dz$$

with canonical native-VC trapezoid ownership. There is no fictitious collapsed-y width. The reported quantities are extensive squared-constraint inventories, not square-root norms or RMS values.

2 Part I: boundary and convergence

2.1 Small-domain localization

At the common terminal time $\tau_c = 3.98614742555601M$, the old full-domain fine-pair orders collapse, while the causally protected interior remains high-order. The terminal full-square values are:

	N128	N256	N512	$p_{128,256}$	$p_{256,512}$
C	5.5801×10^{-2}	3.2813×10^{-3}	1.8063×10^{-3}	4.088	0.861
H	2.1558×10^{-2}	3.5049×10^{-4}	1.2550×10^{-4}	5.943	1.482
M	8.6428×10^{-3}	4.2832×10^{-4}	5.7349×10^{-5}	4.335	2.901
Z	4.0712×10^{-3}	5.0507×10^{-4}	4.0069×10^{-4}	3.011	0.334

Table 1: $R_{\text{out}} = 16M$ full-domain squared-constraint inventories.

Inside $r \leq 4M$, the corresponding $p_{256,512}$ values are 7.831, 7.830, 7.796, and 7.888. For N512, 99.87% of the old Z inventory is outside $r = 12M$: 50.53% in $12 < r \leq 16M$ and 49.34% in the square corners with $r > 16M$.

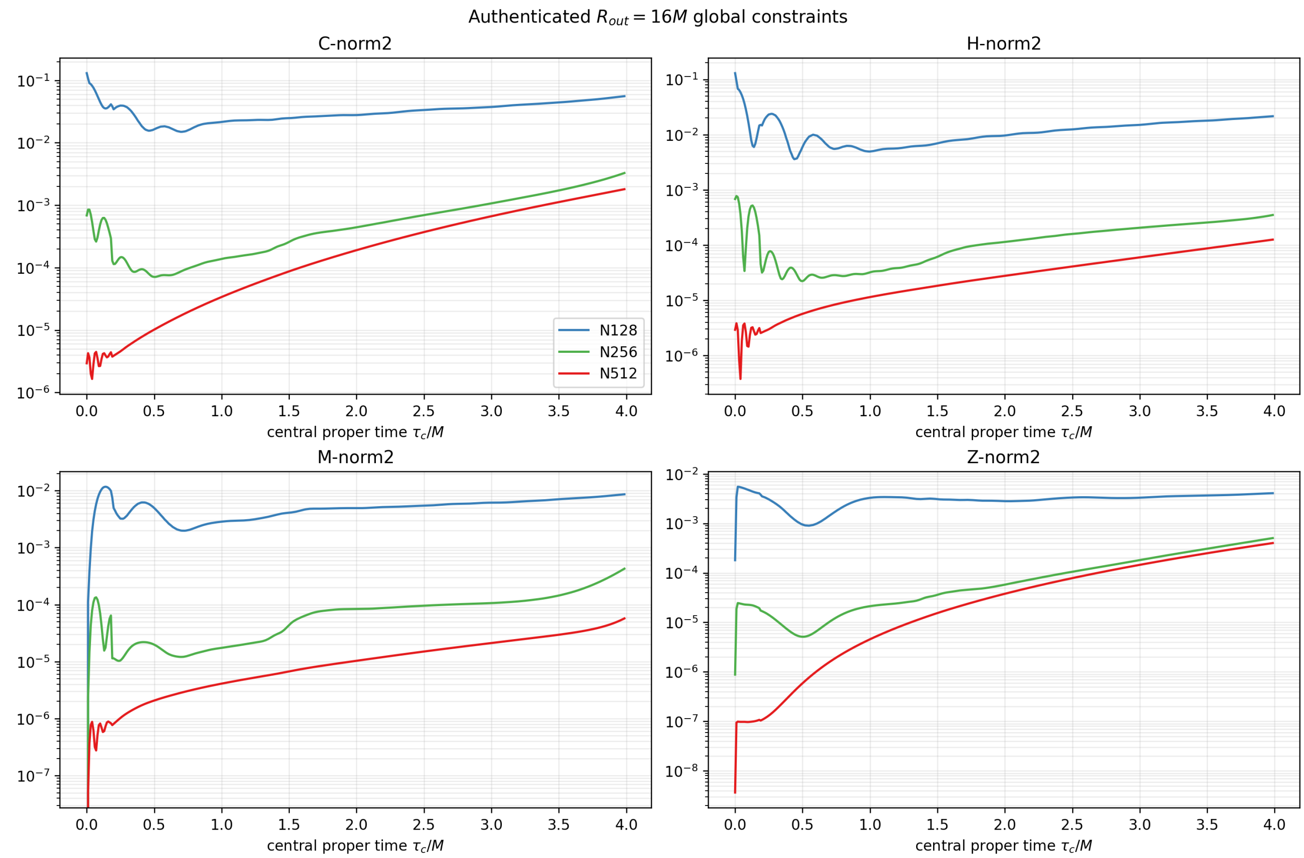

Figure 1: Authenticated $R_{\mathrm{out}} = 16M$ global constraint histories.

2.2 Large-domain construction and authority

The new domain is $\rho \in [0, 128M]$, $z \in [-128M, 128M]$. Three dyadic minimum levels restore exactly the old inner spacings: 0.25, 0.125, and $0.0625M$ for N128/N256/N512. The absolute physical ceiling moves from 20 to 23. In the N256 initialization, 33,153 canonical inner vertices have identical coordinates; 23 of 25 evolved fields agree bitwise and the two Gamma fields differ by at most 1.94×10^{-15}.

The N256 authority ends at $t = 6.5M$ with 206 leaves. A separate N256 replay is bitwise identical in all 264 by 71 history values. N128 and N512 replay every accepted tree exactly. All runs use one A100; N512 peaks at 36,581 MiB HBM.

2.3 Restored convergence

The large-domain full inventories at the common terminal time are:

	N128	N256	N512	$p_{128,256}$	$p_{256,512}$
C	$5.3530{\times}10^{-2}$	$1.1984{\times}10^{-3}$	$3.3856{\times}10^{-5}$	5.481	5.146
H	$2.1422{\times}10^{-2}$	$2.1681{\times}10^{-4}$	$1.9176{\times}10^{-6}$	6.627	6.821
M	$8.6008{\times}10^{-3}$	$3.8660{\times}10^{-4}$	$1.9891{\times}10^{-5}$	4.476	4.281
Z	$3.5586{\times}10^{-3}$	$3.4538{\times}10^{-5}$	$4.7437{\times}10^{-7}$	6.687	6.186

Table 2: $R_{\rm out} = 128M$ full-domain squared-constraint inventories.

The N512 old/new full-domain ratios are 53.35, 65.45, 2.88, and 844.68 for C/H/M/Z. Conversely, $r \leq 4M$ agrees to about one part in 10^9. The maximum small/large difference in $\log_{10}|I(0)|$ is only 6.7×10^{-9} dex.

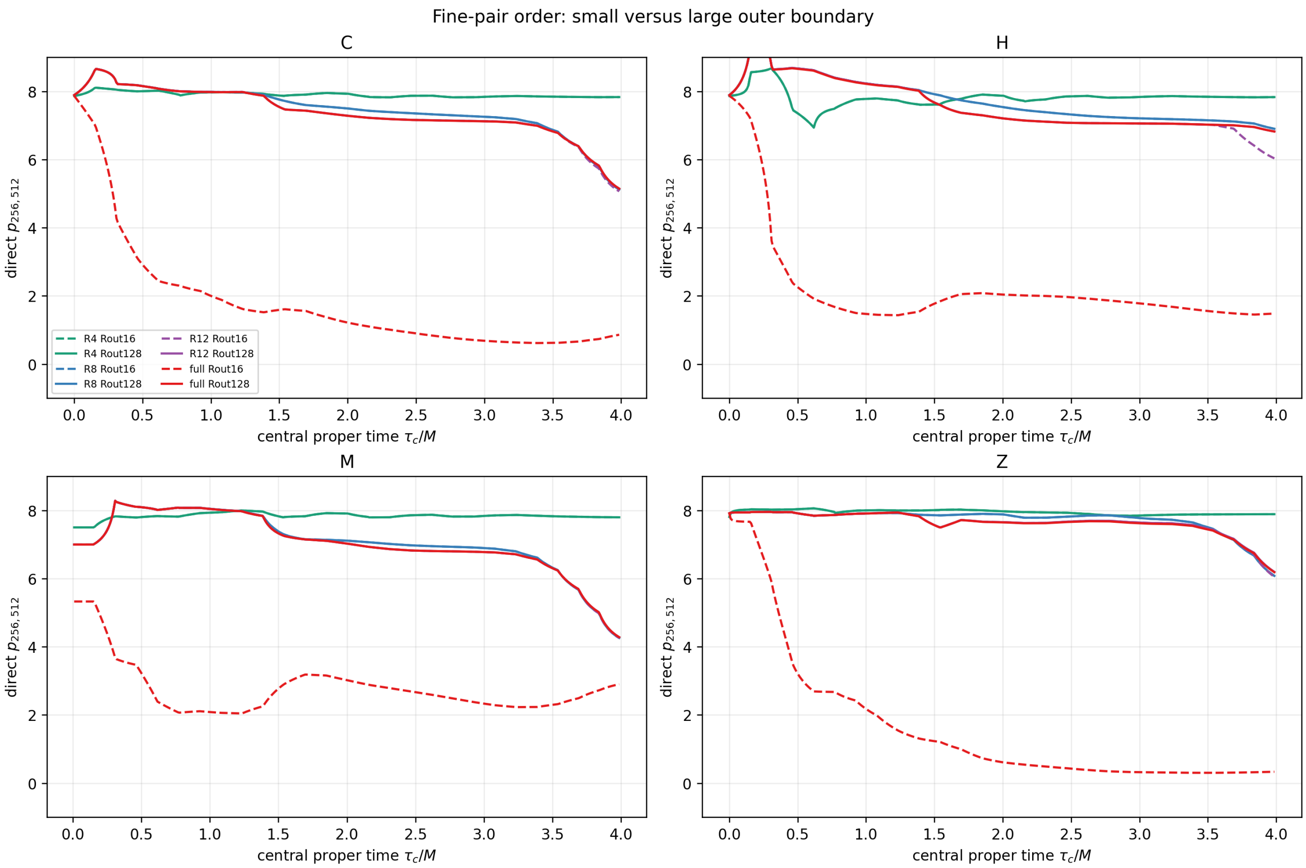

Figure 2: Direct fine-pair order before and after moving the outer boundary.

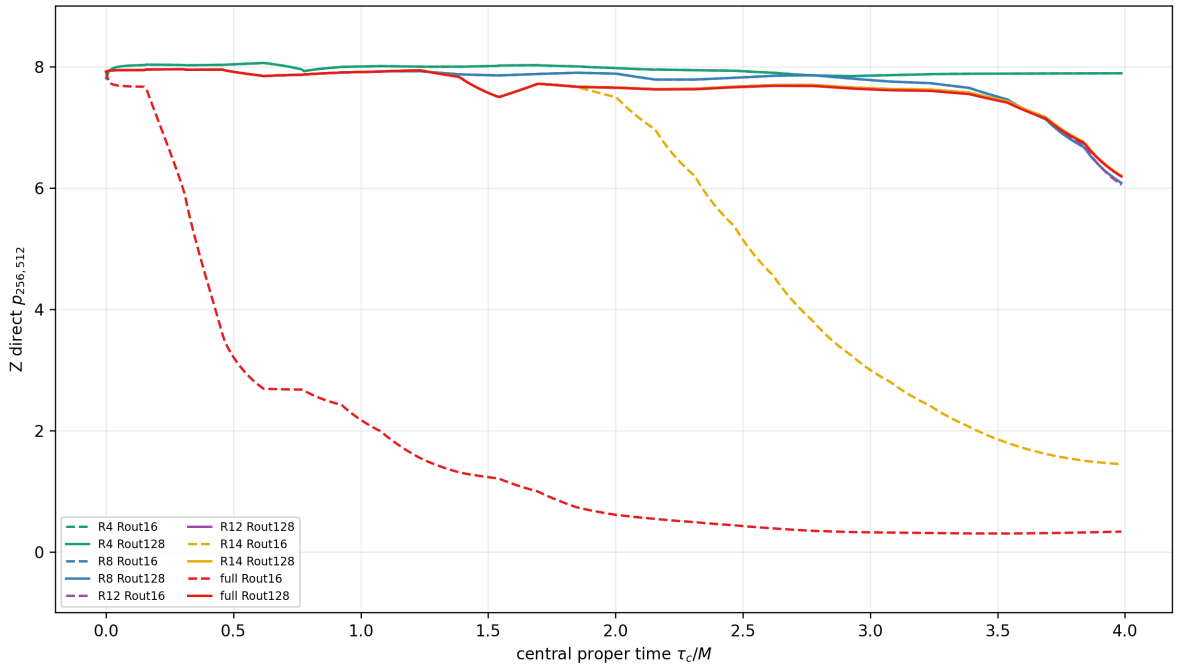

Figure 3: Z convergence identifies the old outer-shell contamination.

Boundary verdict. `BOUNDARY_CONTAMINATION_CONFIRMED`

3 Part II: independent N256 KO scan

Each $R_{\text{out}} = 16M$ case starts from scratch in native-AMR record mode. Only KO dissipation changes.

diss	C(6.5)	C(9.2)	first late refine	max leaves	terminal t	status
0.02	3.285×10^{-3}	1.032	10.2758	1526	11.191917	timestep collapse
0.05	2.693×10^{-3}	0.692	10.3658	65	11.3	reached tlim
0.10	2.224×10^{-3}	0.390	10.7394	86	11.3	reached tlim
0.20	1.850×10^{-3}	0.168	11.1045	50	11.3	reached tlim
0.50	1.518×10^{-3}	0.0437	none	50	11.3	reached tlim

Table 3: Independent N256 KO scan.

The baseline reaches $dt = 7.98 \times 10^{-9}M$, C=$1.61 \times 10^{16}$, physical level 20, and 1,526 leaves before bounded termination. Diss=0.50 has no new late accepted refinement and ends with C=0.2695. The maximum baseline-relative central-trace deviation is 0.028, 0.056, 0.086, and 0.110 dex for diss 0.05–0.50 respectively.

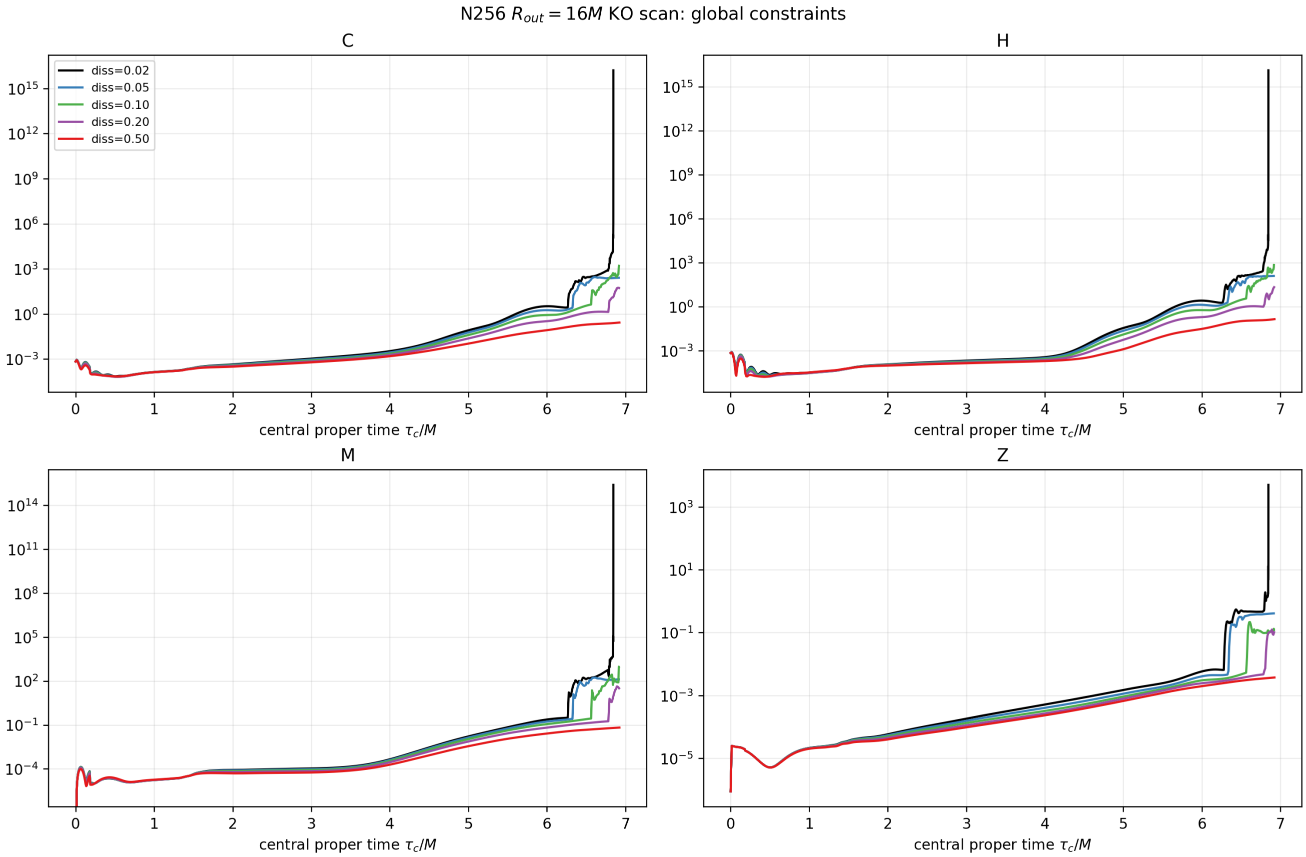

Figure 4: Global constraint inventories versus central proper time.

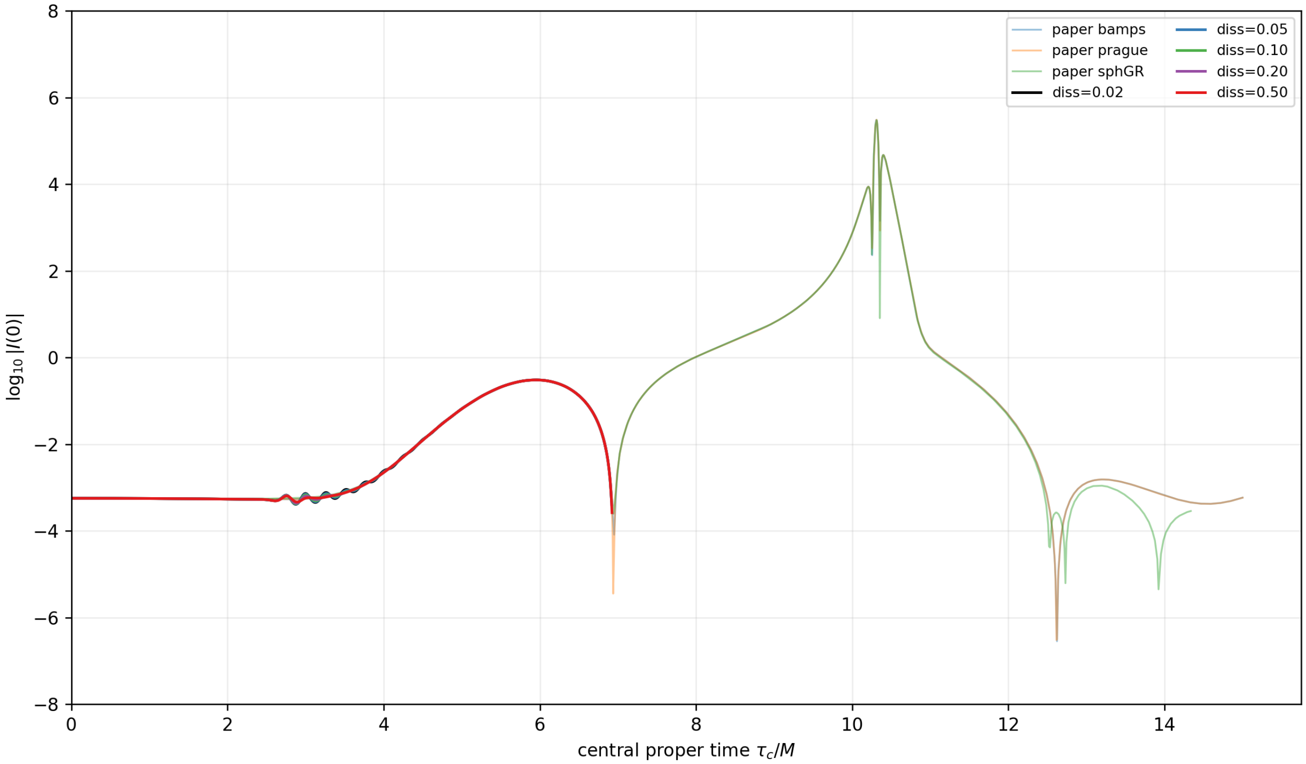

Figure 5: Central curvature trace and authenticated Figure-3 curves, without shifts or rescaling.

KO verdict. `KO_STRONG_EFFECT`

The late cascade contains a strongly under-damped numerical component. The evidence does not justify `KO_OVERDAMPED`: even diss=0.50 does not materially displace the central trace over the matched interval. It also does not justify a production default from one resolution.

4 Synthesis and next campaign

The Figure-2 flattening is primarily an outer-boundary effect. The separate late N256 cascade is strongly sensitive to KO dissipation and therefore is not a KO-insensitive formulation failure.

The best conservative next configuration is the $R_{\text{out}} = 128M$ SMR common-tree workflow with zero Z4 damping and diss=0.20, retaining diss=0.50 as a stability control. A three-resolution extension must show whether the remaining diss=0.20 growth converges away and whether the small diss=0.50 central-trace difference shrinks. No production numerical parameter is changed by this report.

5 Limitations

The boundary qualification ends at $t = 6.5M$; the optional late large-boundary N256 control was not run. The KO study is one-resolution evidence. No late-time Figure-3 convergence, horizon, critical exponent, or continuum stability claim is made. Multi-gigabyte state/restart files remain at the exact Perlmutter paths recorded in the evidence manifest.